Number Theory — problem 6.120

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$a, b, c, d, e \in \mathbb{N}$ are such that $a^4 + b^4 + c^4 + d^4 = e^4$. Show that among the five variables (a) at least three are even, (b) at least three are multiples of 5, (c) at least two are multiples of 10.